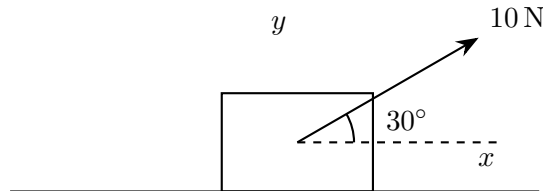


Forces Test (part deux)

Multiple Choice: Please select the best answer choice for each of the following questions.

Questions #1–4 refer to the diagram below.

A 10 N force is applied to a box at an angle of 30 degrees above the horizontal.



1. What is the vertical component of this force?
 - a. 5 N
 - b. 8.7 N
 - c. 10 N
 - d. 5.8 N
2. How does the normal force relate to the force of gravity in this scenario?
 - a. $F_N = F_g$
 - b. $F_N > F_g$
 - c. $F_N < F_g$
 - d. The relationship cannot be determined with the information provided.
3. If the angle of the force is increased, what will happen to the horizontal component?
 - a. It will increase.
 - b. It will decrease.
 - c. It will remain the same.
 - d. This relationship cannot be determined with the information provided.
4. If the angle of the force is increased, what will happen to the normal force?
 - a. It will increase
 - b. It will decrease
 - c. It will remain the same.
 - d. This relationship cannot be determined with the information provided.

$$\Sigma F_x = ma \quad \Sigma F_y = ma \quad F_g = mg \quad F_{fk} = \mu_k F_N \quad 0 < F_{fs} \leq \mu_s F_N$$

5. Which type of friction acts on a book resting on a table?
 - a. Static friction
 - b. Kinetic friction
 - c. Rolling friction
 - d. Fluid friction

6. When an object is placed on a frictionless ramp, what are the two primary forces acting on it?
 - a. Normal force and friction
 - b. Gravity and friction
 - c. Gravity and normal force
 - d. Air resistance and gravity

7. How does the angle of inclination of a ramp affect the horizontal component of the gravitational force (F_{gx})?
 - a. As it increases, the F_{gx} also increases.
 - b. As it increases, the F_{gx} decreases.
 - c. It has no effect on F_{gx}
 - d. As it increases, first it increases F_{gx} , then decreases it.

8. A chair is at rest on the floor ($F_N = F_g$). Bob sits on the chair- how does this affect the relationship between F_N and F_g ?
 - a. F_N decreases because Bob is so skinny.
 - b. F_N increases to support the added force of Bob on the chair.
 - c. F_N remains the same, it can't be bothered with Bob and his shenanigans.
 - d. This cannot be determined without knowing Bob's mood.

$$\Sigma F_x = ma \quad \Sigma F_y = ma \quad F_g = mg \quad F_{fk} = \mu_k F_N \quad 0 < F_{fs} \leq \mu_s F_N$$

Problem Solving

1. A 12 kg block is placed on a frictionless ramp inclined at an angle of 25 degrees above the horizontal.
 - a. Draw a free body diagram for the scenario.
 - b. Write your force summations.
 - c. Determine the acceleration of the block down the ramp.
2. A 10 kg block is pushed across a rough horizontal surface. The coefficient of kinetic friction between the block and the surface is 0.2. If the push force is 50 N:
 - a. Draw a free body diagram for the scenario.
 - b. Determine the magnitude of the normal force acting on the block.
 - c. Determine the force of kinetic friction acting on the block.
 - d. Write your force summations.
 - e. Determine the acceleration of the block.

3. A 25 kg block is resting on a horizontal surface with a coefficient of static friction of 0.4 and a coefficient of kinetic friction of 0.3.
- Determine the maximum friction force.
 - You apply a push force of 90 N: determine the force of friction acting on the block.
 - You apply a push force of 100 N: determine the force of friction acting on the block.
 - Draw the free body diagram for when the object is not moving.
 - If you continue to apply this force, determine the acceleration of the block as it slides across the floor.
4. A 30 kg block is placed on a ramp that has friction inclined at an angle of 30 degrees above the horizontal with a coefficient of static friction of 0.3 and a coefficient of kinetic friction of 0.2.
- Determine F_g , F_{gx} , F_{gy} .
 - You apply a push force of 200 N: determine the force of friction acting on the block.
 - You apply a push force of 400 N: determine the force of friction acting on the block.

- d. Draw the free body diagram for both scenarios.
- e. If you continue to apply this force, determine the acceleration of the block as it slides across the floor.

$$\Sigma F_x = ma \quad \Sigma F_y = ma \quad F_g = mg \quad F_{fk} = \mu_k F_N \quad 0 < F_{fs} \leq \mu_s F_N$$